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CRP-YOLO26: A Lightweight Improved YOLO26 Algorithm for Small Object Detection in UAV Images

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***Abstract*—Small object detection in unmanned aerial vehicle (UAV) imagery remains a challenging task due to the small object scale, complex backgrounds, dense object distribution, and the limited computational resources of edge devices. To address these challenges, this paper proposes CRP-YOLO26, a lightweight improved YOLO26 algorithm specifically designed for UAV small object detection. As a new-generation end-to-end real-time object detector, YOLO26 features NMS-free inference and is well suited for edge deployment. However, when applied to UAV imagery, shallow feature details are easily lost during repeated downsampling, while complex backgrounds reduce the network's ability to focus on target regions.**

To overcome these limitations, a CBAM-Residual attention module is first introduced into the feature fusion network. By jointly exploiting channel attention and spatial attention, the proposed module enhances the representation of informative regions for small objects while preserving original feature details through residual connections. Furthermore, a lightweight P2 detection head is constructed to perform object prediction on higher-resolution shallow feature maps, thereby improving the recall of extremely small objects.

Experiments conducted on a UAV object detection dataset demonstrate that the proposed method effectively improves small object detection accuracy while maintaining low model complexity and computational cost. Compared with the original YOLO26, CRP-YOLO26 achieves a 3.1% improvement in mAP@0.5 and a 4.6% increase in AP for small objects, demonstrating the effectiveness of the proposed approach.

Keywords—YOLO26; UAV detection; small object detection; attention mechanism; lightweight network

I. INTRODUCTION

In recent years, UAV-based vision systems have been widely applied in traffic surveillance, disaster response, agricultural inspection, urban management, and military reconnaissance. Compared with fixed cameras, UAVs provide greater flexibility, wider coverage, and lower deployment costs. However, due to high-altitude imaging and continuously changing viewpoints, objects such as vehicles, pedestrians, and ships often appear with small scales, weak textures, dense distributions, and complex backgrounds, posing significant challenges to real-time object detection[1].

The YOLO series has become one of the most representative real-time object detection frameworks because of its high inference speed, simple architecture, and ease of deployment. As the latest generation of the Ultralytics YOLO family, YOLO26 is specifically designed for edge devices and low-power applications, featuring end-to-end prediction, NMS-free inference, and efficient deployment. Although YOLO26 achieves competitive performance on general object detection benchmarks, it still suffers from two major limitations when directly applied to UAV small object detection[2]. First, repeated downsampling significantly weakens the feature representation of small objects. Second, the complex aerial background often causes the detector to focus on irrelevant regions such as roads, buildings, and vegetation, resulting in missed detections and false positives.

To address these issues, this paper proposes CRP-YOLO26, where CRP denotes the integration of a CBAM-Residual attention module and a P2 detection head. The proposed framework consists of these two complementary improvements, which enhance feature representation while preserving the lightweight characteristics of YOLO26.

The technical novelty of CRP-YOLO26 differs from existing YOLO-based UAV detectors in three aspects:

1.Instead of directly inserting a conventional CBAM module, a lightweight CBAM-Residual module is embedded into the Neck feature fusion stage, which enhances discriminative feature representation while preserving shallow spatial information through residual learning.

2.Unlike conventional P3-P5 detection architectures, a lightweight P2 prediction branch is introduced and cooperates with the proposed attention module, enabling simultaneous enhancement of feature representation and localization for extremely small UAV targets.

3.The proposed design maintains the lightweight characteristics of YOLO26 with only a slight increase in parameters and FLOPs, making it suitable for edge deployment.

II. RELATED WORK

A. UAV Small Object Detection

UAV images are characterized by large fields of view, small object sizes, and complex backgrounds. Traditional object detection methods rely heavily on handcrafted features and therefore struggle to adapt to complex environments. Deep learning-based detectors, such as Faster R-CNN, SSD, and the YOLO series, have significantly improved detection performance[3]. Nevertheless, detecting small objects in UAV imagery remains challenging. Although deep semantic features

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provide strong representational capability, their low spatial resolution often leads to inaccurate localization and missed detections. Therefore, preserving shallow feature details while enhancing the representation of small objects has become a key issue in UAV object detection.[4]

Compared with conventional ground-view images, UAV images exhibit much larger variations in object scale. Objects belonging to the same category may differ substantially due to variations in flight altitude, viewing angle, and object motion. For instance, nearby vehicles usually occupy a relatively large image area with clear structural details, whereas distant vehicles may contain only a few pixels. Such large-scale variations require detection models to possess both robust semantic understanding and fine-grained localization capability. Furthermore, dense object distribution, mutual occlusion, and motion blur commonly observed in UAV imagery further increase the difficulty of object detection[5].

Existing studies generally improve UAV small object detection from three perspectives. The first enhances feature fusion by introducing high-resolution detection layers or improved feature pyramid structures to better combine shallow details with deep semantic information. The second improves the learning capability for small objects through optimized sample assignment strategies, loss functions, or data augmentation techniques. The third incorporates attention mechanisms or contextual modeling modules to enable the network to extract more discriminative target features from complex backgrounds. Although these methods improve detection accuracy, many of them introduce additional model parameters and computational overhead, making deployment on UAV edge devices more difficult. Therefore, this work aims to improve small object detection performance while maintaining model lightweightness and real-time inference capability[6].

B. YOLO26 Object Detection Algorithm

YOLO26 is a new-generation real-time object detection framework designed for efficient end-to-end object detection. Similar to previous YOLO detectors, YOLO26 follows a typical one-stage detection pipeline consisting of three major components: backbone, neck, and detection head. The backbone is responsible for extracting hierarchical visual features from the input image. Through repeated convolution and downsampling operations, the backbone gradually transforms the input image into multi-level feature maps with different spatial resolutions and semantic representations. Shallow layers retain more detailed spatial information such as edges, textures, and object boundaries, whereas deep layers provide stronger semantic information for object classification and localization.

The neck of YOLO26 is used to aggregate multi-scale features from different stages of the backbone. By combining high-level semantic features with low-level spatial details, the neck improves the detection ability for objects with different scales. In the original YOLO26 architecture, the feature pyramid mainly outputs P3, P4, and P5 feature maps for detection. When the input resolution is 640×640, the spatial resolutions of P3, P4, and P5 are typically 80×80, 40×40, and 20×20, respectively. These feature maps are suitable for general object detection. However, for UAV images, extremely small targets may occupy only a few pixels. After repeated downsampling, their feature responses become very weak, which may lead to missed detections.

The detection head of YOLO26 performs object classification and bounding box regression based on the multi-scale feature maps generated by the neck. Different from earlier YOLO variants that rely on Non-Maximum Suppression (NMS) during post-processing, YOLO26 adopts an end-to-end detection design with NMS-free inference. This design reduces the dependence on hand-crafted post-processing and improves inference efficiency. In addition, the detection head adopts a decoupled prediction structure, where classification and localization are optimized through separate branches. This helps reduce the conflict between classification and regression tasks and improves the stability of model training.

During training, YOLO26 optimizes the detector using a combination of classification loss and bounding box regression loss. The classification loss guides the model to distinguish different object categories, while the localization loss constrains the predicted bounding boxes to match the ground-truth boxes more accurately. Meanwhile, an advanced label assignment strategy is used to match predicted candidates with ground-truth objects, enabling the model to learn more stable positive and negative samples during end-to-end optimization. This training mechanism allows YOLO26 to maintain high detection accuracy while reducing redundant predictions.

Compared with earlier YOLO-based detectors, YOLO26 is more suitable for UAV edge deployment for three reasons. First, its lightweight backbone and efficient feature aggregation structure reduce the number of parameters and computational cost. Second, the NMS-free inference pipeline simplifies post-processing and improves real-time performance. Third, the end-to-end detection mechanism makes the model easier to deploy on edge devices with limited computing resources. Therefore, YOLO26 provides a suitable baseline for UAV object detection tasks that require both high accuracy and fast inference.

However, when directly applied to UAV small object detection, the original YOLO26 still has two limitations. First, the P3–P5 detection structure may be insufficient for extremely small objects because shallow spatial details are weakened after downsampling. Second, complex aerial backgrounds such as roads, buildings, trees, and shadows may interfere with feature representation. To address these problems, this paper improves YOLO26 from two aspects: introducing a CBAM-Residual attention module into the feature fusion stage and adding a lightweight P2 detection head for high-resolution small object prediction.

C. Attention Mechanism

Attention mechanisms enable neural networks to focus on more informative spatial regions or feature channels, thereby improving feature representation. For UAV imagery, attention mechanisms can effectively enhance the responses of small target regions while suppressing interference from complex backgrounds. Among existing attention modules, the Convolutional Block Attention Module (CBAM) is a lightweight and effective design that consists of channel attention and spatial attention. It recalibrates feature responses from two complementary perspectives: determining which

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feature channels are more important and which spatial locations deserve greater attention[7]. Considering that UAV small object detection requires both discriminative semantic information and accurate spatial localization, this paper integrates a CBAM-Residual attention module into the feature fusion stage of YOLO26 to enhance feature representation while preserving original feature information.

According to the attention dimension, existing attention mechanisms can generally be categorized into channel attention, spatial attention, and hybrid attention. Channel attention models the importance of different feature channels, enabling the network to emphasize features that are more discriminative for object classification. Spatial attention, on the other hand, focuses on identifying informative regions within an image, allowing the detector to better localize target objects under complex backgrounds. Hybrid attention combines both channel-wise and spatial-wise information, thereby providing stronger feature representation capability. For UAV small object detection, spatial sensitivity is particularly critical because small objects occupy only a limited number of pixels. If the network fails to accurately focus on the target region, the subsequent detection head is unlikely to produce reliable predictions[8].

Although attention mechanisms generally improve detection performance, not all attention modules are suitable for lightweight object detection networks. Some sophisticated attention designs introduce additional matrix operations or computationally expensive convolutions, resulting in increased inference latency[9]. Since UAV platforms are typically constrained by computational resources, power consumption, and memory capacity, lightweight and easily deployable attention modules are more desirable. Therefore, the proposed CRP-YOLO26 incorporates the CBAM-Residual attention module into the feature fusion stage, where multi-scale feature maps are recalibrated in both the channel and spatial dimensions before being fed into the detection head. Furthermore, residual connections are employed to alleviate the over-suppression of shallow feature details caused by attention weighting, thereby preserving fine-grained information that is essential for small object detection[10].

III. Method

A. Overall Architecture

The proposed CRP-YOLO26 is developed based on the original YOLO26 architecture. As illustrated in Fig. 1, the overall framework consists of three components: the Backbone, Neck, and Detection Head. To improve the detection performance of small objects while preserving the lightweight characteristics of YOLO26, this work focuses on enhancing the Neck and Detection Head, without modifying the backbone network.

First, a CBAM-Residual attention module is incorporated into the multi-scale feature fusion stage of the Neck. By jointly exploiting channel attention and spatial attention, the proposed module enhances the network's ability to focus on informative small-object regions in UAV images while preserving shallow feature details through residual connections. Consequently, richer semantic information and more discriminative spatial features are obtained before feature aggregation.

Second, a lightweight P2 detection branch is introduced into the Detection Head. Unlike the original YOLO26, which performs detection using deeper feature maps, the proposed P2 branch utilizes higher-resolution shallow features to generate predictions. This design effectively preserves fine-grained spatial information and improves the localization and classification performance of extremely small objects.

By combining the CBAM-Residual attention module with the P2 detection branch, CRP-YOLO26 simultaneously enhances feature representation and small-object perception while maintaining the lightweight and real-time characteristics of the original YOLO26.

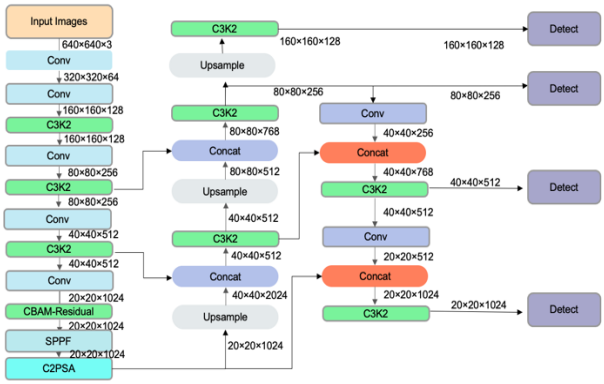


Fig. 1. Overall architecture of the proposed CRP-YOLO26.

B. CBAM-Residual Attention Mechanism

In UAV aerial images, small targets typically occupy only a very small number of pixels and are highly susceptible to interference from complex backgrounds such as buildings, roads, and trees [7]. To enhance the model's attention to small target regions, this paper introduces the CBAM-Residual attention module into the feature fusion network of YOLO26.

Specifically, the proposed CBAM-Residual module is inserted after each multi-scale feature fusion operation in the Neck, immediately following the feature concatenation layer and before the subsequent convolution block. At this stage, feature maps from different resolutions have already been aggregated, allowing the attention module to recalibrate both semantic and spatial information before they are forwarded to the detection head. This design enables the network to suppress background interference while preserving informative shallow features for subsequent prediction.

For an input image of 640 × 640 pixels, the proposed module processes the fused feature maps at multiple scales in the Neck, including P3 (80 × 80), P4 (40 × 40), and P5 (20 × 20). Although these feature maps have different spatial resolutions, they are processed using the same CBAM-Residual structure. The attention module adaptively recalibrates channel responses and spatial importance at each scale, enabling more effective feature representation for objects of different sizes.

This module is built upon the Convolutional Block Attention Module (CBAM) and incorporates residual connections. The proposed module processes the input feature map through four sequential steps:

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The input feature map $F$is first fed into the channel attention branch to estimate the importance of each feature channel.

The channel-refined feature map is then passed to the spatial attention branch to highlight informative spatial regions while suppressing irrelevant background areas.

The attention-weighted feature is obtained by element-wise multiplication with the original feature map.

Finally, a residual shortcut is added to combine the recalibrated feature with the original input, yielding the final output while preserving fine-grained spatial details.

The channel attention module aggregates global contextual information by applying both global average pooling and global max pooling across the spatial dimensions. The pooled features are fed into a shared multi-layer perceptron (MLP), and the channel attention weights are generated via a sigmoid activation:

$$M_c(F) = \sigma(\mathrm{MLP}(\mathrm{AvgPool}(F)) + \mathrm{MLP}(\mathrm{MaxPool}(F)))$$

The spatial attention module further emphasizes informative spatial locations by concatenating the average-pooled and max-pooled feature maps along the channel dimension, followed by a 7×7convolution and sigmoid activation:

$$M_s(F) = \sigma(f^{7\times7}([\mathrm{AvgPool}(F); \mathrm{MaxPool}(F)]))$$

The two attention maps are then applied sequentially to the input feature map. The final attention-enhanced feature is computed as:

$$F_{\mathrm{cbam}} = F \otimes (M_c(F) \otimes M_s(F))$$

where $\otimes$denotes element-wise multiplication. To prevent the attention mechanism from overly suppressing shallow-layer detail information (such as edges, textures, and positional cues), a residual connection is introduced:

$$F_{\mathrm{out}} = F_{\mathrm{cbam}} + F$$

Channel attention highlights the feature channels that are more discriminative for target classification, while spatial attention enhances the response in regions where small targets are likely to appear. The residual connection allows the network to strengthen important target responses while effectively preserving the expressive power of the original features.

This design is particularly suitable for UAV-based small target detection, where targets often occupy only a few pixels and rely heavily on low-level texture, edge, and positional information. Without the residual pathway, excessive attention suppression could easily lead to the loss of critical target details. The residual structure effectively mitigates this issue and improves overall detection performance.

The proposed CBAM-Residual module introduces only lightweight pooling, shared MLP, and a lightweight spatial convolution. Therefore, the additional computational overhead is negligible compared with the overall network, making it suitable for lightweight UAV deployment.

The architecture of the CBAM-Residual model is shown in Fig. 2.

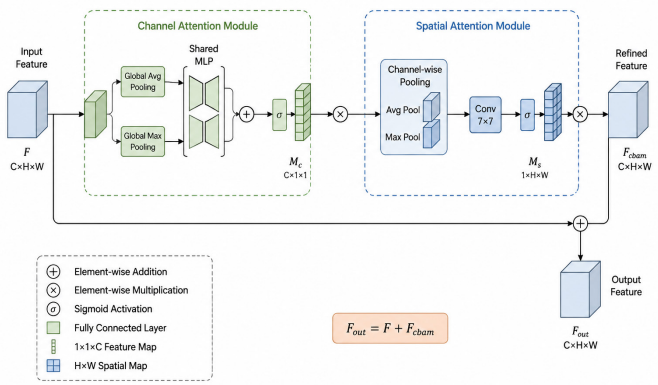


Fig. 2. Structure of the CBAM-Residual module.

C. *Lightweight P2 Small Target Detection Head*

The YOLO series detection networks typically use multi-scale feature layers such as P3, P4, and P5 for prediction. When the input image resolution is 640×640, the typical sizes of each feature map are: P3 layer 80×80, P4 layer 40×40, and P5 layer 20×20 [8]. For extremely small targets in UAV aerial images that contain only a dozen or so pixels, after 8× or higher downsampling, they often correspond to only a few grid cells on the feature map. This results in extremely weak feature representation and significantly increases detection difficulty.

To improve the model's detection capability for extremely small targets, this paper introduces a P2 detection head. The P2 feature map has a smaller downsampling factor and higher spatial resolution, which better preserves edge, texture, and positional information in shallow features, effectively alleviating the problem of small target feature degradation [9].

To avoid a significant increase in model computational cost caused by adding the P2 branch, this paper adopts a lightweight convolutional structure to build the P2 detection head, thereby improving detection performance while minimizing the number of parameters and FLOPs as much as possible [10].

The multi-scale prediction process of the P2 detection head can be expressed as:

$$Y = \mathrm{Concat}(\mathrm{Head}_{P2}(F_{P2}), \mathrm{Head}_{P3}(F_{P3}), \mathrm{Head}_{P4}(F_{P4}), \mathrm{Head}_{P5}(F_{P5}))$$

where F_{P2}, F_{P3}, F_{P4}, and F_{P5}represent the feature maps of the corresponding scales, and Head_{Pi}denotes the detection head at that scale. The different detection heads have clear divisions of labor: the P2 branch is primarily responsible for extremely small target detection, the P3 branch for small target detection, and the P4 and P5 branches focus on medium-to-large target detection.

By adding the high-resolution P2 detection branch, the model can more fully utilize shallow detail information, significantly improving its perception and localization capability for extremely small targets in UAV images.

Different from previous methods that independently introduce attention modules or additional detection heads, the proposed CRP-YOLO26 jointly optimizes feature enhancement and shallow feature perception. The CBAM-Residual module suppresses background interference while preserving original information through residual learning, whereas the P2 detection

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head complements the preserved shallow features for extremely small object localization.

IV. Experiment Design And Result Analysis

A. Dataset Introduction

The experiments in this paper utilize a publicly available UAV aerial image dataset released on the Kaggle platform for training and testing. This dataset was collected by UAVs at different altitudes, viewing angles, and under complex environmental conditions. It includes various scene targets such as buildings, roads, vehicles, and pedestrians. The dataset features large variations in target scales, complex backgrounds, and a high proportion of small targets, making it suitable for effectively validating the practical application performance of object detection algorithms in UAV scenarios.

The dataset contains 3 categories and is divided into training, validation, and test sets. The training set contains 10,799 images, the validation set contains 603 images, and the test set contains 596 images. All images have a uniform resolution of 640×640. They are annotated using a consistent labeling format and stored according to the data organization requirements of the YOLO object detection framework to ensure experimental reproducibility and fairness.

In UAV aerial images, targets are usually far from the camera, resulting in them occupying only a small pixel area in the image. In particular, vehicles, pedestrians, and other targets often exhibit characteristics such as small size, low resolution, and blurred edges. Meanwhile, building textures, road markings, and shadow areas in complex backgrounds can easily interfere with target recognition. Therefore, small target detection in UAV scenarios has always been an important research direction in the field of object detection.

To verify the improved model's detection capability for small targets, the experimental analysis further evaluates the detection performance across targets of different scales. Compared with medium and large targets, small targets have insufficient feature information, strong context dependency, and high sensitivity to localization errors, making them more prone to missed detections and false detections. Consequently, small target detection performance can more directly reflect the model's practical application value in UAV aerial photography scenarios. Dataset samples are shown in Fig.3:

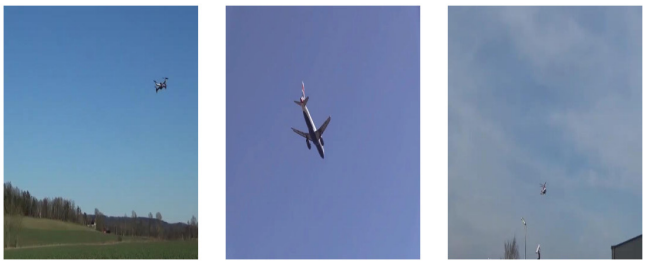


Fig. 3. Representative samples from the UAV image dataset.

B. Experimental Details

The experiments in this paper were conducted on an Ubuntu 20.04 operating system, using an NVIDIA RTX 3090 GPU as the hardware platform and PyTorch as the deep learning framework. During training, the input image size was uniformly set to 640×640, the number of training epochs was set to 300, and the batch size was set to 16. The model was trained using the official default optimization strategy of YOLO26, with an initial learning rate of 0.01.

To improve the model's robustness and generalization ability, data augmentation strategies such as Mosaic, Random Scaling, Color Jitter, and Random Flip were adopted during the training phase to enhance the model's adaptability to targets of different scales and complex scenes. To ensure the fairness and comparability of experimental results, all comparative models were trained and tested using the same dataset split, input size, and training parameters.

This paper uses Precision, Recall, mAP@0.5, mAP@0.5:0.95, number of parameters (Parameters), and floating-point operations (FLOPs) as evaluation metrics. Among them, Precision measures the proportion of true targets among the samples predicted as positive by the model, while Recall measures the proportion of true targets that are correctly detected. mAP@0.5 represents the mean average precision across all categories at an IoU threshold of 0.5. mAP@0.5:0.95 comprehensively evaluates detection performance across multiple IoU thresholds and can more fully reflect the model's overall detection capability.

In addition, considering the large number of distant and small-sized targets in UAV aerial photography scenarios, this paper further focuses on the model's performance in small target detection tasks and analyzes the improvement effects of the proposed methods on small target feature extraction and localization capabilities in combination with the detection results.

C. Comparative Experiments

To verify the effectiveness of the proposed CRP-YOLO26 model, this paper selects YOLOv8n, YOLO11n, and YOLO26n as baseline models for comparison. All models were trained and evaluated using the same dataset split, training parameters, and test environment to ensure the fairness of the experimental results. Table 1 presents the detection performance comparison of different models on the Kaggle Drone Dataset.

TABLE I. COMPARISON OF DETECTION PERFORMANCE OF DIFFERENT ALGORITHMS

Method	Precision /%	Recall/ %	mAP@0.5 /%	mAP@0.5:0.95/%	Parameters/M	FLOPs /G
YOLOv8n	92.1	90.7	92.8	81.5	3.2	8.7
YOLO11n	92.8	91.4	93.5	82.8	2.6	6.5
YOLO26n	93.4	92.1	94.2	84.1	2.8	6.9
CRP-YOLO26	95.1	93.8	95.8	86.7	3.1	7.6

As shown in Table 1, the proposed CRP-YOLO26 achieves the best detection performance across all evaluation metrics. This indicates that the proposed improvement strategies can effectively enhance the model's target recognition capability and localization accuracy.

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In terms of Precision and Recall metrics, CRP-YOLO26 shows significant improvements, demonstrating that the model not only improves detection accuracy but also effectively reduces missed detections. This is mainly attributed to the P2 small target detection head's full utilization of shallow fine-grained features and the CBAM-Residual attention module's enhancement of key target regions, enabling the model to more accurately extract target features and suppress interference from complex backgrounds.

Regarding model complexity, the parameters of CRP-YOLO26 only increased from 2.8M to 3.1M, and FLOPs increased from 6.9G to 7.6G, representing a relatively small growth. Compared with the improvement in detection performance, the additional computational overhead is acceptable. This demonstrates that the proposed method maintains good lightweight characteristics and deployment potential while ensuring high detection accuracy.

Although the proposed model introduces a slight increase in parameters and FLOPs, the performance gain in UAV small object detection significantly outweighs the additional computational cost, demonstrating that the proposed improvements remain suitable for real-time edge deployment.

The comprehensive experimental results show that the proposed CRP-YOLO26 can achieve superior detection performance at a relatively low computational cost, demonstrating strong application value in UAV aerial target detection tasks.

D. Ablation Experiments

To verify the contribution of each improved module to the model performance, this paper uses YOLO26n as the baseline model. Ablation experiments were conducted by introducing the CBAM-Residual attention module and the P2 small target detection head separately, and then constructing the complete CRP-YOLO26 model. All experiments used the same dataset split, training parameters, and test environment to ensure fairness and comparability of the results. The experimental results are shown in Table 2.

TABLE II. ABLATION EXPERIMENT RESULTS

Method	CBAM-Residual	P2 Detection Head	Precision/%	Recall/%	mAP@0.5/%	mAP@0.5:0.95/%	Parameters/M
YOLO26n	×	×	93.4	92.1	94.2	84.1	2.8
YOLO26n + CBAM-Residual	√	×	94.0	92.8	94.9	85.0	2.9
YOLO26n + P2	×	√	94.4	93.2	95.3	85.8	3.0
CRP-YOLO26	√	√	95.1	93.8	95.8	86.7	3.1

As shown in Table 2, after introducing the CBAM-Residual attention module alone, the model's mAP@0.5 increased from 94.2% to 94.9%, and mAP@0.5:0.95 increased from 84.1% to 85.0%. This indicates that the module can effectively enhance the network's attention to key target regions. At the same time, the residual connection helps retain the original feature information, thereby improving target feature representation and detection accuracy.

When only the P2 small target detection head is introduced, the model's mAP@0.5 improves to 95.3% and mAP@0.5:0.95 improves to 85.8%, showing a larger gain than adding the attention module alone. This demonstrates that high-resolution shallow features play a crucial role in small-scale target detection in UAV aerial photography scenarios. The improvement brought by the P2 detection head is mainly reflected in extremely small objects. Since the P2 branch utilizes higher-resolution shallow feature maps, more spatial details and boundary information can be preserved before repeated downsampling. Consequently, the localization accuracy and recall of small objects are significantly improved. In contrast, medium-sized and large objects are already well represented by the deeper P3–P5 feature maps. Therefore, the introduction of the P2 branch has only a limited influence on their detection performance while maintaining the original detection capability.

When both the CBAM-Residual attention module and the P2 small target detection head are introduced simultaneously, CRP-YOLO26 achieves the best results across all evaluation metrics. The experimental results indicate that the two improved modules exhibit good synergistic effects: the CBAM-Residual module strengthens target feature representation, while the P2 detection head enhances the utilization of shallow detail information. Together, they significantly improve the model's detection performance in UAV aerial photography scenarios.

E. Detection Results

To intuitively demonstrate the detection performance of the proposed method, visualization comparisons were conducted on distant dense vehicle areas, occluded pedestrian areas, and complex road background areas. CRP-YOLO26 is able to detect more distant small targets and maintain more complete detection results in densely populated target regions. Specific detection results are shown in Figure 4:



Fig 4. Detection results on the UAV dataset.

From the visualization results, it can be seen that CRP-YOLO26 can detect more small-scale targets with more accurate bounding boxes. This indicates that the P2 detection head

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effectively preserves shallow spatial information, while the CBAM-Residual attention module further enhances the response in target regions.

V. Conclusion

This paper proposes a lightweight improved YOLO26 algorithm, CRP-YOLO26, for small target detection in UAV images. To address the problems of weak small target features, complex backgrounds, and information loss due to downsampling in UAV images, this paper introduces the CBAM-Residual attention mechanism and the P2 small target detection head, respectively. The former enhances the model's attention to key target regions through channel and spatial attention while using residual connections to preserve original detailed features. Experimental analysis further demonstrates that the introduced P2 detection head mainly benefits extremely small object detection while preserving the detection performance of medium-sized and large objects. Compared with existing lightweight YOLO-based UAV detectors, the proposed CRP-YOLO26 achieves a better trade-off between detection accuracy and computational efficiency, demonstrating its practical value for UAV edge deployment.

Experimental results show that CRP-YOLO26 outperforms the original YOLO26 in detection accuracy and small target recall while maintaining a relatively low number of parameters and computational cost. Future work will further investigate model pruning, quantization deployment, and the robustness of UAV target detection under complex weather conditions.

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